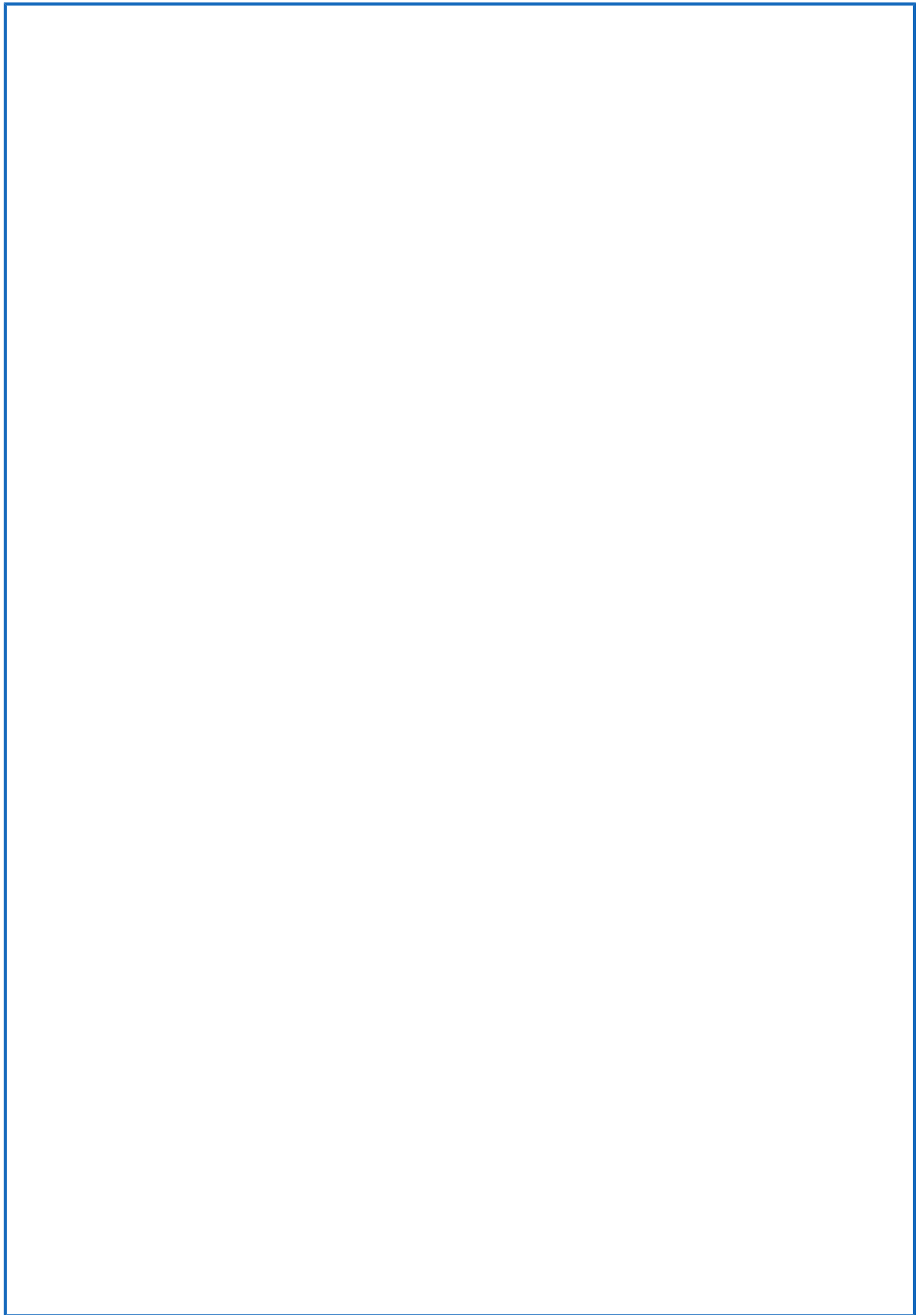
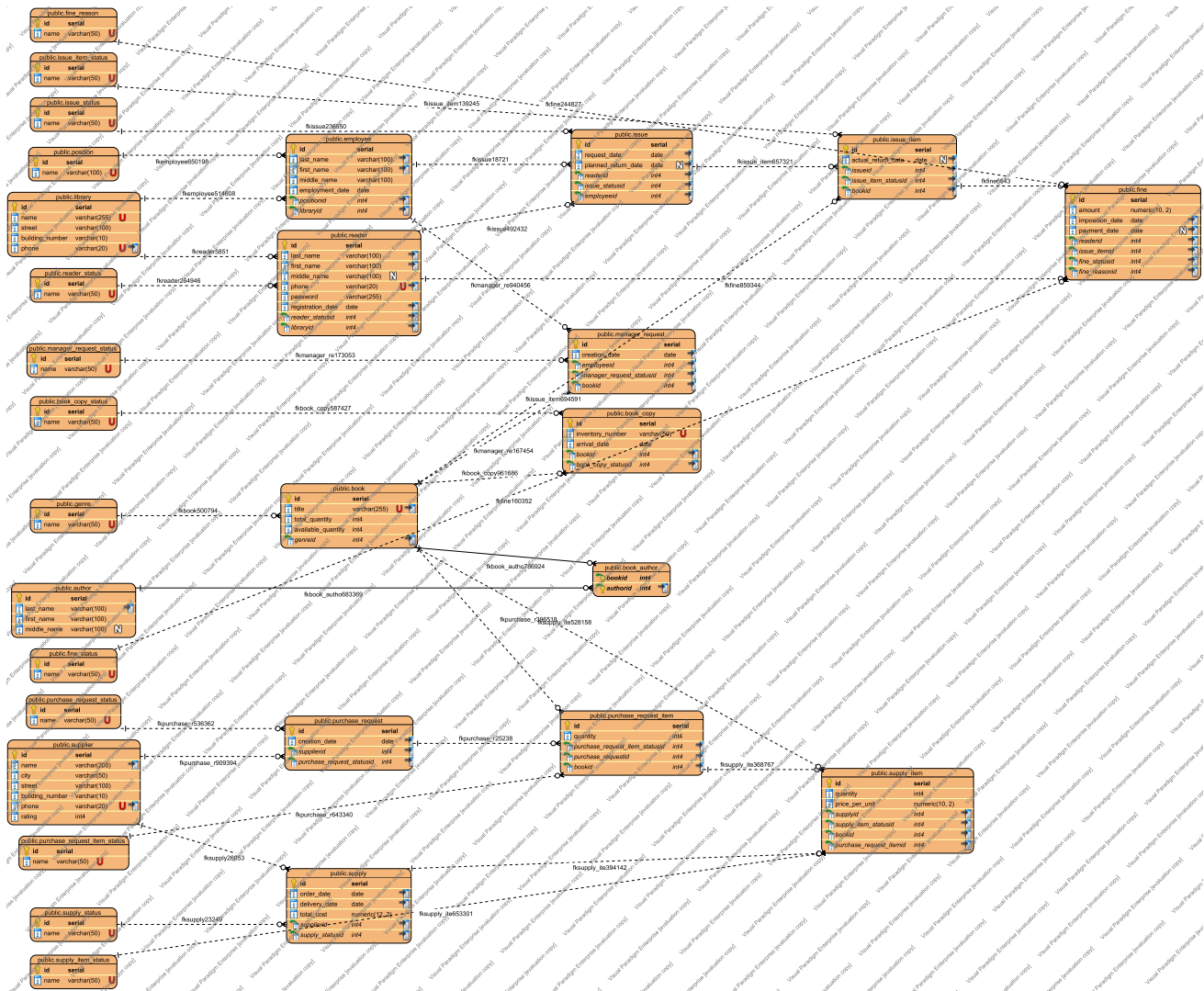




1. Entity Relationship Diagram1



1.1. author

1.1.1. Properties

Discriminator	N/A
Class model	N/A
Schema	public

1.1.2. Column Summary

Name	Description
first_name	
id	
last_name	
middle_name	

1.1.3. Column Details

first_name

Type	varchar
Length	100
Scale	0

id

Type	int4
Length	0
Scale	0
Primary Key	true

last_name

Type	varchar
Length	100
Scale	0

middle_name

Type	varchar
Length	100
Scale	0
Nullable	true

1.1.4. Indices

Name	Column
idx_author_last_name	last_name

1.1.5. Relationships

Relationship	From	To
 fkbook_autho683369	 author	 book_author

1.2. book

1.2.1. Properties

Discriminator	N/A
Class model	N/A
Schema	public

1.2.2. Column Summary

Name	Description
 available_quantity	
 genreid	
 id	

Name	Description
 title	
 total_quantity	

1.2.3. Column Details

available_quantity

Type	int4
Length	0
Scale	0

genreid

Type	int4
Length	0
Scale	0
Foreign Key	true

id

Type	int4
Length	0
Scale	0
Primary Key	true

title

Type	varchar
Length	255
Scale	0

total_quantity

Type	int4
Length	0
Scale	0

1.2.4. Indices

Name	Column
idx_book_genreid	genreid
idx_book_title	title

1.2.5. Constraints

Name	Type	Constraint
book_title_key	Unique	UNIQUE(title)

1.2.6. Relationships

Relationship	From	To
 fkbook_autho786924	 book	 book_author
 fkbook_copy961686	 book	 book_copy
 fkbook500794	 genre	 book
 fkissue_item694591	 book	 issue_item
 fkmanager_re167454	 book	 manager_request
 fkpurchase_r396518	 book	 purchase_request_item
 fksupply_ite528158	 book	 supply_item

1.3. book_author

1.3.1. Properties

Discriminator	N/A
Class model	N/A
Schema	public

1.3.2. Column Summary

Name	Description
 authorid	
 bookid	

1.3.3. Column Details

authorid

Type	int4
Length	0
Scale	0
Primary Key	true
Foreign Key	true

bookid

Type	int4
Length	0
Scale	0
Primary Key	true
Foreign Key	true

1.3.4. Indices

Name	Column
idx_book_author_authorid	authorid

1.3.5. Relationships

Relationship	From	To
 fkbook_autho683369	 author	 book_author
 fkbook_autho786924	 book	 book_author

1.4. book_copy

1.4.1. Properties

Discriminator	N/A
Class model	N/A
Schema	public

1.4.2. Column Summary

Name	Description
 arrival_date	
 book_copy_statusid	
 bookid	
 id	
 inventory_number	

1.4.3. Column Details

arrival_date

Type	date
Length	0
Scale	0

book_copy_statusid

Type	int4
Length	0
Scale	0
Foreign Key	true

bookid

Type	int4
Length	0
Scale	0
Foreign Key	true

id

Type	int4
Length	0

Scale	0
Primary Key	true
 <i>inventory_number</i>	
Type	varchar
Length	50
Scale	0

1.4.4. Indices

Name	Column
idx_book_copy_bookid	bookid
idx_book_copy_statusid	book_copy_statusid

1.4.5. Constraints

Name	Type	Constraint
book_copy_inventory_number_key	Unique	UNIQUE(inventory_number)

1.4.6. Relationships

Relationship	From	To
 fkbook_copy587427	 book_copy_status	 book_copy
 fkbook_copy961686	 book	 book_copy

1.5. book_copy_status

1.5.1. Properties

Discriminator	N/A
Class model	N/A
Schema	public

1.5.2. Column Summary

Name	Description
 id	
 name	

1.5.3. Column Details

 *id*

Type	int4
Length	0
Scale	0
Primary Key	true
 <i>name</i>	
Type	varchar

Length 50

Scale 0

1.5.4. Constraints

Name	Type	Constraint
book_copy_status_name_key	Unique	UNIQUE(name)

1.5.5. Relationships

Relationship	From	To
 fkbook_copy587427	 book_copy_status	 book_copy

1.6. employee

1.6.1. Properties

Discriminator	N/A
Class model	N/A
Schema	public

1.6.2. Column Summary

Name	Description
 employment_date	
 first_name	
 id	
 last_name	
 libraryid	
 middle_name	
 positionid	

1.6.3. Column Details

employment_date

Type	date
Length	0
Scale	0

first_name

Type	varchar
Length	100
Scale	0

id

Type	int4
------	------

Length	0
Scale	0
Primary Key	true
 <i>last_name</i>	
Type	varchar
Length	100
Scale	0
 <i>libraryid</i>	
Type	int4
Length	0
Scale	0
Foreign Key	true
 <i>middle_name</i>	
Type	varchar
Length	100
Scale	0
 <i>positionid</i>	
Type	int4
Length	0
Scale	0
Foreign Key	true

1.6.4. Indices

Name	Column
idx_employee_last_name	last_name
idx_employee_libraryid	libraryid
idx_employee_name_search	first_name, last_name
idx_employee_positionid	positionid

1.6.5. Relationships

Relationship	From	To
 fkemployee514608	 library	 employee
 fkemployee650198	 position	 employee
 fkissue18721	 employee	 issue
 fkmanager_re940456	 employee	 manager_request

1.7. fine

1.7.1. Properties

Discriminator	N/A
Class model	N/A
Schema	public

1.7.2. Column Summary

Name	Description
 amount	
 fine_reasonid	
 fine_statusid	
 id	
 imposition_date	
 issue_itemid	
 payment_date	
 readerid	

1.7.3. Column Details

amount

Type	numeric
Length	10
Scale	2

fine_reasonid

Type	int4
Length	0
Scale	0
Foreign Key	true

fine_statusid

Type	int4
Length	0
Scale	0
Foreign Key	true

id

Type	int4
Length	0
Scale	0
Primary Key	true

imposition_date

Type	date
Length	0
Scale	0

 [issue_itemid](#)

Type	int4
Length	0
Scale	0
Foreign Key	true

 [payment_date](#)

Type	date
Length	0
Scale	0
Nullable	true

 [readerid](#)

Type	int4
Length	0
Scale	0
Foreign Key	true

1.7.4. Indices

Name	Column
idx_fine_dates	imposition_date, payment_date
idx_fine_imposition_date	imposition_date
idx_fine_issue_itemid	issue_itemid
idx_fine_payment_date	payment_date
idx_fine_readerid	readerid
idx_fine_reasonid	fine_reasonid
idx_fine_statusid	fine_statusid

1.7.5. Relationships

Relationship	From	To
 fkfine160352	 fine_status	 fine
 fkfine244827	 fine_reason	 fine
 fkfine6543	 issue_item	 fine
 fkfine859344	 reader	 fine

1.8. fine_reason

1.8.1. Properties

Discriminator	N/A
Class model	N/A
Schema	public

1.8.2. Column Summary

Name	Description
 id	
 name	

1.8.3. Column Details

 *id*

Type	int4
Length	0
Scale	0
Primary Key	true
 <i>name</i>	
Type	varchar
Length	50
Scale	0

1.8.4. Constraints

Name	Type	Constraint
fine_reason_name_key	Unique	UNIQUE(name)

1.8.5. Relationships

Relationship	From	To
 fine244827	 fine_reason	 fine

1.9. fine_status

1.9.1. Properties

Discriminator	N/A
Class model	N/A
Schema	public

1.9.2. Column Summary

Name	Description
 id	
 name	

1.9.3. Column Details

 *id*

Type	int4
Length	0
Scale	0
Primary Key	true

 *name*

Type	varchar
Length	50
Scale	0

1.9.4. Constraints

Name	Type	Constraint
fine_status_name_key	Unique	UNIQUE(name)

1.9.5. Relationships

Relationship	From	To
 fine160352	 fine_status	 fine

1.10. genre

1.10.1. Properties

Discriminator	N/A
Class model	N/A
Schema	public

1.10.2. Column Summary

Name	Description
 id	
 name	

1.10.3. Column Details

 *id*

Type	int4
Length	0
Scale	0
Primary Key	true

 *name*

Type	varchar
Length	50
Scale	0

1.10.4. Constraints

Name	Type	Constraint
genre_name_key	Unique	UNIQUE(name)

1.10.5. Relationships

Relationship	From	To
 fkbook500794	 genre	 book

1.11. issue

1.11.1. Properties

Discriminator	N/A
Class model	N/A
Schema	public

1.11.2. Column Summary

Name	Description
 employeeid	
 id	
 issue_statusid	
 planned_return_date	
 readerid	
 request_date	

1.11.3. Column Details

employeeid

Type	int4
Length	0
Scale	0
Foreign Key	true

id

Type	int4
Length	0
Scale	0
Primary Key	true

issue_statusid

Type	int4
Length	0

Scale	0
Foreign Key	true
 planned_return_date	
Type	date
Length	0
Scale	0
Nullable	true

 [readerid](#)

Type	int4
Length	0
Scale	0
Foreign Key	true

 [request_date](#)

Type	date
Length	0
Scale	0

1.11.4. Indices

Name	Column
idx_issue_dates	planned_return_date, request_date
idx_issue_employeeid	employeeid
idx_issue_planned_return_date	planned_return_date
idx_issue_readerid	readerid
idx_issue_request_date	request_date
idx_issue_statusid	issue_statusid

1.11.5. Relationships

Relationship	From	To
-fkissu657321	 issue	 issue_item
-fkissu18721	 employee	 issue
-fkissu236550	 issue_status	 issue
-fkissu492432	 reader	 issue

1.12. issue_item

1.12.1. Properties

Discriminator	N/A
Class model	N/A
Schema	public

1.12.2. Column Summary

Name	Description
 actual_return_date	
 bookid	
 id	
 issue_item_statusid	
 issueid	

1.12.3. Column Details

actual_return_date

Type	date
Length	0
Scale	0
Nullable	true

bookid

Type	int4
Length	0
Scale	0
Foreign Key	true

id

Type	int4
Length	0
Scale	0
Primary Key	true

issue_item_statusid

Type	int4
Length	0
Scale	0
Foreign Key	true

issueid

Type	int4
Length	0
Scale	0
Foreign Key	true

1.12.4. Indices

Name	Column
idx_issue_item_actual_return_date	actual_return_date
idx_issue_item_bookid	bookid
idx_issue_item_issueid	issueid
idx_issue_item_statusid	issue_item_statusid

1.12.5. Relationships

Relationship	From	To
fkfine6543	 issue_item	 fine
fkissue_item139245	 issue_item_status	 issue_item
fkissue_item657321	 issue	 issue_item
fkissue_item694591	 book	 issue_item

1.13. issue_item_status

1.13.1. Properties

Discriminator	N/A
Class model	N/A
Schema	public

1.13.2. Column Summary

Name	Description
 id	

 [name](#)

1.13.3. Column Details

 [id](#)

Type	int4
Length	0
Scale	0
Primary Key	true

 [name](#)

Type	varchar
Length	50
Scale	0

1.13.4. Constraints

Name	Type	Constraint
issue_item_status_name_key	Unique	UNIQUE(name)

1.13.5. Relationships

Relationship	From	To
 fkissue_item139245	 issue_item_status	 issue_item

1.14. issue_status

1.14.1. Properties

Discriminator	N/A
Class model	N/A
Schema	public

1.14.2. Column Summary

Name	Description
 id	

 name

1.14.3. Column Details

 *id*

Type	int4
Length	0
Scale	0
Primary Key	true

 *name*

Type	varchar
Length	50
Scale	0

1.14.4. Constraints

Name	Type	Constraint
issue_status_name_key	Unique	UNIQUE(name)

1.14.5. Relationships

Relationship	From	To
 fkissue236550	 issue_status	 issue

1.15. library

1.15.1. Properties

Discriminator	N/A
Class model	N/A
Schema	public

1.15.2. Column Summary

Name	Description
 building_number	
 id	
 name	
 phone	
 street	

1.15.3. Column Details

building_number

Type	varchar
Length	10
Scale	0

id

Type	int4
Length	0
Scale	0
Primary Key	true

name

Type	varchar
Length	255
Scale	0

phone

Type	varchar
Length	20
Scale	0

street

Type	varchar
Length	100
Scale	0

1.15.4. Indices

Name	Column
idx_library_phone	phone

1.15.5. Constraints

Name	Type	Constraint
library_name_key	Unique	UNIQUE(name)
library_phone_key	Unique	UNIQUE(phone)

1.15.6. Relationships

Relationship	From	To
 fkemployee514608	 library	 employee
 fkreader5851	 library	 reader

1.16. manager_request

1.16.1. Properties

Discriminator	N/A
Class model	N/A
Schema	public

1.16.2. Column Summary

Name	Description
 bookid	
 creation_date	
 employeeid	
 id	
 manager_request_statusid	

1.16.3. Column Details

bookid

Type	int4
Length	0
Scale	0
Foreign Key	true

creation_date

Type	date
Length	0
Scale	0

employeeid

Type	int4
Length	0
Scale	0

Foreign Key true

 *id*

Type int4

Length 0

Scale 0

Primary Key true

 *manager_request_statusid*

Type int4

Length 0

Scale 0

Foreign Key true

1.16.4. Indices

Name	Column
------	--------

idx_manager_request_bookid	bookid
----------------------------	--------

idx_manager_request_creation_date	creation_date
-----------------------------------	---------------

idx_manager_request_employeeid	employeeid
--------------------------------	------------

idx_manager_request_statusid	manager_request_statusid
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1.16.5. Relationships

Relationship	From	To
 fkmanager_re167454	 <i>book</i>	 <i>manager_request</i>
 fkmanager_re173053	 <i>manager_request_status</i>	 <i>manager_request</i>
 fkmanager_re940456	 <i>employee</i>	 <i>manager_request</i>

1.17. manager_request_status

1.17.1. Properties

Discriminator N/A

Class model N/A

Schema public

1.17.2. Column Summary

Name	Description
------	-------------

 id	
--	--

 name	
--	--

1.17.3. Column Details

 *id*

Type	int4
Length	0
Scale	0
Primary Key	true

 *name*

Type	varchar
Length	50
Scale	0

1.17.4. Constraints

Name	Type	Constraint
manager_request_status_name_key	Unique	UNIQUE(name)

1.17.5. Relationships

Relationship	From	To
 fkmanager_re173053	 manager_request_status	 manager_request

1.18. position

1.18.1. Properties

Discriminator	N/A
Class model	N/A
Schema	public

1.18.2. Column Summary

Name	Description
 id	
 name	

1.18.3. Column Details

 *id*

Type	int4
Length	0
Scale	0
Primary Key	true

 *name*

Type	varchar
Length	100
Scale	0

1.18.4. Constraints

Name	Type	Constraint
position_name_key	Unique	UNIQUE(name)

1.18.5. Relationships

Relationship	From	To
fkemployee650198	 position	 employee

1.19. purchase_request

1.19.1. Properties

Discriminator	N/A
Class model	N/A
Schema	public

1.19.2. Column Summary

Name	Description
 creation_date	
 id	
 purchase_request_statusid	
 supplierid	

1.19.3. Column Details

creation_date

Type	date
Length	0
Scale	0

id

Type	int4
Length	0
Scale	0
Primary Key	true

purchase_request_statusid

Type	int4
Length	0
Scale	0
Foreign Key	true

supplierid

Type	int4
------	------

Length	0
Scale	0
Foreign Key	true

1.19.4. Indices

Name	Column
idx_purchase_request_creation_date	creation_date
idx_purchase_request_status_id	purchase_request_statusid
idx_purchase_request_supplier_id	supplierid

1.19.5. Relationships

Relationship	From	To
fkpurchase_r25238	 purchase_request	 purchase_request_item
fkpurchase_r536362	 purchase_request_status	 purchase_request
fkpurchase_r909394	 supplier	 purchase_request

1.20. purchase_request_item

1.20.1. Properties

Discriminator	N/A
Class model	N/A
Schema	public

1.20.2. Column Summary

Name	Description
 bookid	
 id	
 purchase_request_item_statusid	
 purchase_requestid	
 quantity	

1.20.3. Column Details

 **bookid**

Type	int4
Length	0
Scale	0
Foreign Key	true



id

Type	int4
Length	0
Scale	0
Primary Key	true

purchase_request_item_statusid

Type	int4
Length	0
Scale	0
Foreign Key	true

purchase_requestid

Type	int4
Length	0
Scale	0
Foreign Key	true

quantity

Type	int4
Length	0
Scale	0

1.20.4. Indices

Name	Column
idx_purchase_request_item_bookid	bookid
idx_purchase_request_item_purchase_requestid_requestid	purchase_requestid requestid
idx_purchase_request_item_purchase_request_item_statusid_statusid	purchase_request_item_statusid statusid

1.20.5. Relationships

Relationship	From	To
-fkpurchase_r25238	purchase_request	purchase_request_item
-fkpurchase_r396518	book	purchase_request_item
-fkpurchase_r643340	purchase_request_item_status	purchase_request_item
-fksupply_ite368767	purchase_request_item	supply_item

1.21. purchase_request_item_status

1.21.1. Properties

Discriminator	N/A
---------------	-----

Class model	N/A
Schema	public

1.21.2. Column Summary

Name	Description
 id	
 name	

1.21.3. Column Details

 *id*

Type	int4
Length	0
Scale	0
Primary Key	true

 *name*

Type	varchar
Length	50
Scale	0

1.21.4. Constraints

Name	Type	Constraint
purchase_request_item_status_name_key	Unique	UNIQUE(name)

1.21.5. Relationships

Relationship	From	To
fkpurchase_r643340	 purchase_request_item_status	 purchase_request_item

1.22. purchase_request_status

1.22.1. Properties

Discriminator	N/A
Class model	N/A
Schema	public

1.22.2. Column Summary

Name	Description
 id	
 name	

1.22.3. Column Details

 *id*

Type	int4
Length	0
Scale	0
Primary Key	true

 ***name***

Type	varchar
Length	50
Scale	0

1.22.4. Constraints

Name	Type	Constraint
purchase_request_status_name_key	Unique	UNIQUE(name)

1.22.5. Relationships

Relationship	From	To
 fkpurchase_r536362	 purchase_request_status	 purchase_request

1.23. reader

1.23.1. Properties

Discriminator	N/A
Class model	N/A
Schema	public

1.23.2. Column Summary

Name	Description
 first_name	
 id	
 last_name	
 libraryid	
 middle_name	
 password	
 phone	
 reader_statusid	
 registration_date	

1.23.3. Column Details

 ***first_name***

Type	varchar
------	---------

Length 100

Scale 0

 *id*

Type int4

Length 0

Scale 0

Primary Key true

 *last_name*

Type varchar

Length 100

Scale 0

 *libraryid*

Type int4

Length 0

Scale 0

Foreign Key true

 *middle_name*

Type varchar

Length 100

Scale 0

Nullable true

 *password*

Type varchar

Length 255

Scale 0

 *phone*

Type varchar

Length 20

Scale 0

 *reader_statusid*

Type int4

Length 0

Scale 0

Foreign Key true

 *registration_date*

Type date

Length	0
Scale	0

1.23.4. Indices

Name	Column
idx_reader_last_name	last_name
idx_reader_libraryid	libraryid
idx_reader_name_search	first_name, last_name
idx_reader_phone	phone
idx_reader_registration_date	registration_date
idx_reader_statusid	reader_statusid

1.23.5. Constraints

Name	Type	Constraint
reader_phone_key	Unique	UNIQUE(phone)

1.23.6. Relationships

Relationship	From	To
 fkfine859344	 reader	 fine
 fkissue492432	 reader	 issue
 fkreader264946	 reader_status	 reader
 fkreader5851	 library	 reader

1.24. reader_status

1.24.1. Properties

Discriminator	N/A
Class model	N/A
Schema	public

1.24.2. Column Summary

Name	Description
 id	

 name

1.24.3. Column Details

 *id*

Type	int4
Length	0
Scale	0
Primary Key	true

 name

Type	varchar
Length	50
Scale	0

1.24.4. Constraints

Name	Type	Constraint
reader_status_name_key	Unique	UNIQUE(name)

1.24.5. Relationships

Relationship	From	To
 fkreader264946	 reader_status	 reader

 1.25. supplier

1.25.1. Properties

Discriminator	N/A
Class model	N/A
Schema	public

1.25.2. Column Summary

Name	Description
 building_number	
 city	
 id	
 name	
 phone	
 rating	
 street	

1.25.3. Column Details

 building_number

Type	varchar
Length	10
Scale	0

 city

Type	varchar
Length	50
Scale	0



id

Type	int4
Length	0
Scale	0
Primary Key	true

 **name**

Type	varchar
Length	200
Scale	0

 **phone**

Type	varchar
Length	20
Scale	0

 **rating**

Type	int4
Length	0
Scale	0

 **street**

Type	varchar
Length	100
Scale	0

1.25.4. Indices

Name	Column
idx_supplier_name	name
idx_supplier_phone	phone

1.25.5. Constraints

Name	Type	Constraint
supplier_phone_key	Unique	UNIQUE(phone)

1.25.6. Relationships

Relationship	From	To
 fkpurchase_r909394	 supplier	 purchase_request
 fksupply26053	 supplier	 supply

1.26. supply

1.26.1. Properties

Discriminator	N/A
---------------	-----

Class model	N/A
Schema	public

1.26.2. Column Summary

Name	Description
 delivery_date	
 id	
 order_date	
 supplierid	
 supply_statusid	
 total_cost	

1.26.3. Column Details

 delivery_date

Type	date
Length	0
Scale	0

 *id*

Type	int4
Length	0
Scale	0
Primary Key	true

 order_date

Type	date
Length	0
Scale	0

 supplierid

Type	int4
Length	0
Scale	0
Foreign Key	true

 supply_statusid

Type	int4
Length	0
Scale	0
Foreign Key	true

total_cost

Type	numeric
Length	12
Scale	2

1.26.4. Indices

Name	Column
idx_supply_delivery_date	delivery_date
idx_supply_order_date	order_date
idx_supply_statusid	supply_statusid
idx_supply_supplierid	supplierid

1.26.5. Relationships

Relationship	From	To
 fksupply_ite394142	 supply	 supply_item
 fksupply23249	 supply_status	 supply
 fksupply26053	 supplier	 supply

1.27. supply_item

1.27.1. Properties

Discriminator	N/A
Class model	N/A
Schema	public

1.27.2. Column Summary

Name	Description
 bookid	
 id	
 price_per_unit	
 purchase_request_ itemid	
 quantity	
 supply_item_status id	
 supplyid	

1.27.3. Column Details

bookid

Type	int4
------	------

Length	0
Scale	0
Foreign Key	true
 <i>id</i>	

Type	int4
Length	0
Scale	0
Primary Key	true
 <i>price_per_unit</i>	

Type	numeric
Length	10
Scale	2
 <i>purchase_request_itemid</i>	

Type	int4
Length	0
Scale	0
Foreign Key	true
 <i>quantity</i>	

Type	int4
Length	0
Scale	0
 <i>supply_item_statusid</i>	

Type	int4
Length	0
Scale	0
Foreign Key	true
 <i>supplyid</i>	

Type	int4
Length	0
Scale	0
Foreign Key	true

1.27.4. Indices

Name	Column
idx_supply_item_bookid	bookid
idx_supply_item_request_itemid	purchase_request_itemid
idx_supply_item_statusid	supply_item_statusid

Name	Column
idx_supply_item_supplyid	supplyid

1.27.5. Relationships

Relationship	From	To
fksupply_ite368767	purchase_request_item	supply_item
fksupply_ite394142	supply	supply_item
fksupply_ite528158	book	supply_item
fksupply_ite653301	supply_item_status	supply_item

1.28. supply_item_status

1.28.1. Properties

Discriminator	N/A
Class model	N/A
Schema	public

1.28.2. Column Summary

Name	Description
id	
name	

1.28.3. Column Details

id

Type	int4
Length	0
Scale	0
Primary Key	true

name

Type	varchar
Length	50
Scale	0

1.28.4. Constraints

Name	Type	Constraint
supply_item_status_name_key	Unique	UNIQUE(name)

1.28.5. Relationships

Relationship	From	To
fksupply_ite653301	supply_item_status	supply_item

1.29. supply_status

1.29.1. Properties

Discriminator	N/A
Class model	N/A
Schema	public

1.29.2. Column Summary

Name	Description
 id	
 name	

1.29.3. Column Details

 *id*

Type	int4
Length	0
Scale	0
Primary Key	true

 *name*

Type	varchar
Length	50
Scale	0

1.29.4. Constraints

Name	Type	Constraint
supply_status_name_key	Unique	UNIQUE(name)

1.29.5. Relationships

Relationship	From	To
 fksupply23249	 supply_status	 supply